Detailed Project Proposal (DPP)

Design and Evaluation of an Explainable Machine Learning Prototype for Postpartum Depression Risk Estimation

7COM1040: Computer Science Masters Project

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Aim of the Project

The aim of this project is to explore the potential of machine learning in estimating the risk of postpartum depression among new mothers. The study will examine whether demographic, lifestyle, and psychological factors can be used to identify individuals who may be at increased risk. Different machine learning models will be evaluated to determine their effectiveness, and explainable AI techniques will be used to improve understanding of how predictions are generated. The project will also include a simple mobile prototype to demonstrate how risk estimates and explanatory information can be presented in a clear and user friendly way.

Research gap

Many existing studies on postpartum depression focus on prediction performance, while limited attention is given to explaining how prediction results are produced. There is also a lack of simple systems that present risk estimates and explanations in an understandable way. This project aims to address these limitations by combining machine learning, explainable predictions, and a user-friendly prototype.

Research Question

1. How accurately can machine learning models estimate postpartum depression risk from demographic, lifestyle and psychological variables?
2. Which factors have the greatest influence on postpartum depression risk estimation?

Hypothesis

It is expected that machine learning models can be used to estimate postpartum depression risk using demographic, lifestyle, and psychological factors. It is also expected that explainable techniques will help improve understanding of the prediction results.

Objectives

1. To review existing research on postpartum depression and its associated risk factors.
2. To identify and prepare a suitable public dataset for postpartum depression risk estimation.
3. To develop and compare machine learning models for estimating postpartum depression risk.
4. To evaluate model performance using appropriate classification metrics.
5. To identify the factors that have the greatest influence on prediction outcomes.
6. To develop a simple mobile application prototype that presents risk estimates and explanations in a user-friendly manner.

Short Description of the Idea

Postpartum depression is a mental health condition that affects some women after childbirth and can have a significant impact on their wellbeing. Identifying individuals who may be at risk at an early stage could help improve awareness and encourage timely support.

This project aims to explore the use of machine learning techniques for estimating postpartum depression risk using demographic, lifestyle, and psychological factors. Several machine learning models will be developed and evaluated to determine their effectiveness. The project will also investigate which factors have the greatest influence on risk estimates. To demonstrate the practical application of the proposed approach, a simple mobile prototype will be developed that allows users to enter relevant information and view the estimated risk along with a clear explanation of the results.

Research Methodology

1. Literature Review

   - A review of existing research will be conducted to understand postpartum depression, its risk factors, and current approaches used for risk estimation.
   - Previous studies related to machine learning, explainable models, and digital mental health systems will be examined to identify suitable techniques for this project.

2. Data Collection

   - This study will use the "PostPartum Depression Dataset" available on Kaggle.
   - The dataset contains demographic, lifestyle, and psychological variables that are relevant to postpartum depression risk, including factors such as age, stress levels, sleep quality, emotional wellbeing, and social support.
   - This dataset has been selected because it provides suitable variables for machine learning analysis and is appropriate for evaluating postpartum depression risk estimation models.
   - The collected data will be used for model training, testing, and performance evaluation throughout the study.

3. Data Preparation

- The collected dataset will be examined to identify missing values, duplicate records, and any inconsistencies within the data.
- Appropriate data cleaning techniques will be applied to improve data quality and ensure reliable analysis.
- Relevant features associated with postpartum depression risk will be selected for model development.
- The dataset will be divided into training and testing sets to support model training and performance evaluation.

4. Machine Learning Model Development

- Several machine learning models, including Logistic Regression, Random Forest, and XGBoost, will be implemented and compared.
- The models will be trained to estimate postpartum depression risk using the selected variables.
- Model parameters may be adjusted to improve performance.

5. Explainable Analysis

- SHAP will be used to investigate how different features influence model predictions.
- Both overall feature importance and individual explanations will be examined.

6. Prototype Development

- A simple mobile application prototype will be developed to demonstrate the prediction process.
- The prototype will allow users to enter information and view risk estimates together with explanatory information.

7. Evaluation

- Model performance will be evaluated using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and confusion matrices.
- The performance of different models will be compared to identify the most suitable approach.
- The usability and functionality of the prototype will also be reviewed.

8. Ethical Considerations

- The project will use publicly available data and will not collect personal information from real users.
- The system will be presented as a research prototype and not as a medical diagnostic tool.
- Privacy, fairness, and potential limitations of the model will be discussed.

9. Expected contribution

- The project is expected to contribute an explainable machine learning prototype for postpartum depression risk estimation. It will compare different machine learning models, identify important risk factors, and demonstrate how prediction results can be presented through a simple and accessible user interface

10. Risks and fallback plan

- One potential risk is that the selected dataset may contain limited or incomplete information, which could affect model performance. To address this, alternative public datasets will be considered if required. Another risk is that the machine learning models may not achieve the expected performance. In such cases, different preprocessing techniques and model configurations will be explored. If there are time constraints during development, the project will priorities the machine learning evaluation and explanation components, while ensuring that a functional prototype is available for demonstration.

References

1. T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2016.
2. S. M. Lundberg and S. I. Lee, "A Unified Approach to Interpreting Model Predictions," in Advances in Neural Information Processing Systems, 2017.
3. S. Shorey et al., "Prevalence and Incidence of Postpartum Depression Among Healthy Mothers: A Systematic Review and Meta-Analysis," Journal of Psychiatric Research, vol. 104, pp. 235–248, 2018.